Lesson notes · for you

Repeat it: loops

Programming · Level 1 · Lesson 2

Time
20 min

Materials
Paper and a pencil; Arrow cards and number cards 1 to 4; A phone (optional)

Learning goals

- Spot steps that repeat.
- Use a loop to make a program shorter.
- Count cards to find the shortest program.

Activity: Clap the loop

1. Warm-up (3 min): Clap “clap, clap, stamp” four times. How would you tell someone to do it, in as few words as possible?
2. Write (7 min): Write the pattern in full (12 moves), then as a loop: “clap, clap, stamp: 4 times”. Count the words in each.
3. Create (7 min): Make up your own pattern of three moves and write it with a loop.
4. Share (3 min): Give your loop to a partner. Can they perform it exactly?

No device is needed: hands, feet and paper are enough.

Discussion

- Why is the loop easier to change?
- Which songs or games have loops?
- When would you not use a loop?

Check yourself

- Write one thing you do again and again as a loop.

Try it this week
Teach someone at home a clapping pattern using only a loop: “clap, clap, stamp: 3 times”. Did they do it right?

Worksheet

Repeat it: loops

Programming · Level 1 · Lesson 2

Name _____ Class _____ Date _____

1 **Which program is shorter, but walks the same way?**

◯ They are the same length ◯ Forward, forward, forward ◯ Forward, three times

2 **Walk the robot to the flag with 5 cards at most. Without loops you would need 9!**

					★
→					

Cut out cards (↑ forward, ↶ turn left, ↷ turn right) and lay out a program that takes the robot to the ★. Write it here:

3 **Match each long program to its loop.**

forward, forward, forward, forward	• • • •	turn right × 2
turn right, turn right	• • • •	jump × 3
jump, jump, jump	• • • •	forward × 4

4 **A program: forward × 3, turn right, forward × 2. How many cards does it use?**

◯ 5 ◯ 6 ◯ 3

5 **"Turn right × 4" leaves the robot facing the same way as when it started.**

◯ True ◯ False

✎ **Find a loop in your day: something you do again and again. How many times do you repeat it?**

Check your answers

Repeat it: loops

Programming · Level 1 · Lesson 2

1 **Forward, three times**

One card that repeats is a loop: it is shorter, and the robot walks exactly the same way.

2 **Any program that takes the robot to the ★ with 5 cards or fewer.**

Forward 4 times, forward 1 more time, turn left, forward 3 times: four cards.

3 **forward, forward, forward, forward — forward × 4**
turn right, turn right — turn right × 2
jump, jump, jump — jump × 3

Count how many times the same step comes, and write it once with that number: that is a loop.

4 **3**

Three cards: forward (3 times), turn right, forward (2 times).

5 **True**

Four quarter-turns make a whole turn: up, right, down, left, and back to up.